

# CAMPUS PLACEMENT ASSESSMENT 2026

National College of Engineering & Technology - Training & Placement Cell

Duration: 30 Minutes | Total Marks: 10 | Negative Marking: 0.25

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## SECTION A: QUANTITATIVE APTITUDE

1. A train running at 54 km/hr crosses a platform 150 meters long in 20 seconds. What is the length of the train in meters?

- (A) 120 meters
- (B) 150 meters
- (C) 180 meters
- (D) 200 meters

Answer: B |  $\text{Speed} = 54 \times (5/18) = 15 \text{ m/s}$ . Distance in 20s =  $15 \times 20 = 300\text{m}$ . Train length =  $300 - 150 = 150 \text{ meters}$ .

2. If 12 men can complete a project in 15 days, how many men are needed to complete the same project in 10 days?

- (A) 16 men
- (B) 18 men
- (C) 20 men
- (D) 22 men

Answer: B |  $M1 \times D1 = M2 \times D2 \Rightarrow 12 \times 15 = M2 \times 10 \Rightarrow 180 = 10 \times M2 \Rightarrow M2 = 18$ .

3. Find the missing number in the sequence: 4, 9, 19, 39, 79, ?

- (A) 149
- (B) 159
- (C) 169
- (D) 179

Answer: B | Pattern:  $(\times 2 + 1)$ .  $4 \times 2 + 1 = 9$ ,  $9 \times 2 + 1 = 19$ ,  $19 \times 2 + 1 = 39$ ,  $39 \times 2 + 1 = 79$ ,  $79 \times 2 + 1 = 159$ .

## SECTION B: DATA STRUCTURES & ALGORITHMS

4. What is the worst-case time complexity of searching an element in a balanced Binary Search Tree (AVL Tree)?

- (A)  $O(1)$
- (B)  $O(\log n)$
- (C)  $O(n)$
- (D)  $O(n \log n)$

Answer: B | A balanced BST maintains height of  $O(\log n)$ , so search is  $O(\log n)$  even in worst case.

5. Which data structure is primarily used to implement Breadth-First Search (BFS) graph traversal?

- (A) Stack
- (B) Priority Queue
- (C) Queue
- (D) Hash Table

Answer: C | BFS processes nodes level by level using FIFO order, which is implemented with a Queue.

6. What is the auxiliary space complexity of standard in-place Quicksort algorithm?

- (A)  $O(1)$
- (B)  $O(\log n)$
- (C)  $O(n)$
- (D)  $O(n^2)$

Answer: B | Due to recursive call stack on partition, auxiliary space is  $O(\log n)$  on average.

7. In a max-heap of n elements, what is the time complexity to insert a new element?

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- (A)  $O(1)$
- (B)  $O(\log n)$
- (C)  $O(n)$
- (D)  $O(n \log n)$

Answer: B | Inserting an element appends to bottom and heapifies up along the tree height, taking  $O(\log n)$ .

## SECTION C: DATABASE & CORE COMPUTING

**8. Which normal form eliminates partial functional dependency on the primary key?**

- (A) First Normal Form (1NF)
- (B) Second Normal Form (2NF)
- (C) Third Normal Form (3NF)
- (D) Boyce-Codd Normal Form (BCNF)

Answer: B | 2NF requires 1NF and no non-prime attribute should be functionally dependent on part of candidate key.

**9. Which HTTP status code corresponds to 'Unauthorized' access?**

- (A) 400
- (B) 401
- (C) 403
- (D) 404

Answer: B | 401 stands for Unauthorized (unauthenticated), whereas 403 is Forbidden.

**10. Which of the following ACID properties ensures that database transactions are safely persisted even in case of power failure?**

- (A) Atomicity
- (B) Consistency
- (C) Isolation
- (D) Durability

Answer: D | Durability guarantees that once a transaction commits, it remains committed even after crash or power failure.

## ANSWER KEY SUMMARY:

1: B 2: B 3: B 4: B 5: C 6: B 7: B 8: B 9: B 10: D